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Redefining Deepwater Limits Through Cross-Border Synergy: The Field A Case Study

Nurul Izza B. Rashid, Budi Mawardi, A. Fakrudin Zakeria, Fazrul A. Zawawi, N. Asyikin M. Radzuan, and M. Aminuddin Md Karim, Malaysian Petroleum Management, PETRONAS, Malaysia

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Abstract

The 4th phase of the Field A development project marks a strategic expansion of production from the South-West sector of this prolific deepwater field spanning offshore Malaysia and Brunei. Operated by Operator G, a multinational oil and gas company on behalf of nine joint venture partners, the field is subject to the regulatory oversight of both Malaysia and Brunei. Following the Unitization Agreement (UA) and the Unit Operating Agreement (UOA), the project aims to enhance hydrocarbon recovery through three new oil producers and one water injector, targeting undeveloped reservoir. The project is designed to improve sweep efficiency and increase cumulative recovery, which has already exceeded 350 MMstb since first oil. Malaysia Petroleum Management (MPM), the custodian of Malaysia's upstream petroleum resources, empowered to act on behalf of PETRONAS in this strategic management and stewardship of the nation's hydrocarbon assets.

MPM's role as a regulator, shaper, and accelerator is tasked to ensure that the project is executed efficiently, complies with regulations, and delivers maximum value to all stakeholders. This regulatory oversight is crucial for the smooth operation and success of the project, while ensuring environmental protection through technology advancement and enhance security aligned. Key challenges during the operation include subsurface uncertainty, extreme extended reach drilling (EERD) trajectories, and unprecedented cross-border coordination. To address these challenges, a comprehensive framework has been established to ensure alignment among all parties involved. Stakeholder mapping and process opportunities helps maintain clear communication and collaboration between operators, Malaysia, and Brunei. Besides addressing the challenges, this alignment is essential for seizing cost optimization opportunities and ensuring that the project is executed efficiently, delivering maximum value to all stakeholders. This strategic importance underscores the critical role of regulatory oversight, commercial mindset, and stakeholder alignment in achieving successful and sustainable outcomes.

Introduction and Background

The era of deepwater development in Malaysia began with a significant breakthrough in August 2007, when the first deepwater project achieved its first oil, just five years after the initial discovery in the

field. These deepwater discoveries have shifted the focus towards deepwater exploration and positioned Malaysia, through PETRONAS, as a highly promising deepwater hub for the region. Building on this momentum, Malaysia has since ventured into cross-border deepwater collaborations with neighbouring country, leveraging its technical expertise and strategic location to jointly develop transboundary resources and unlock shared hydrocarbon potential

Field A is a deepwater oil field located 120 km offshore northeastern of Sabah, with water depths between 2800 and 4000 ft. Field A straddles the border between Malaysia and Brunei waters consist of 4 blocks as shown in Figure 1. The fields are unitized through a UA signed in 2021 by both governments

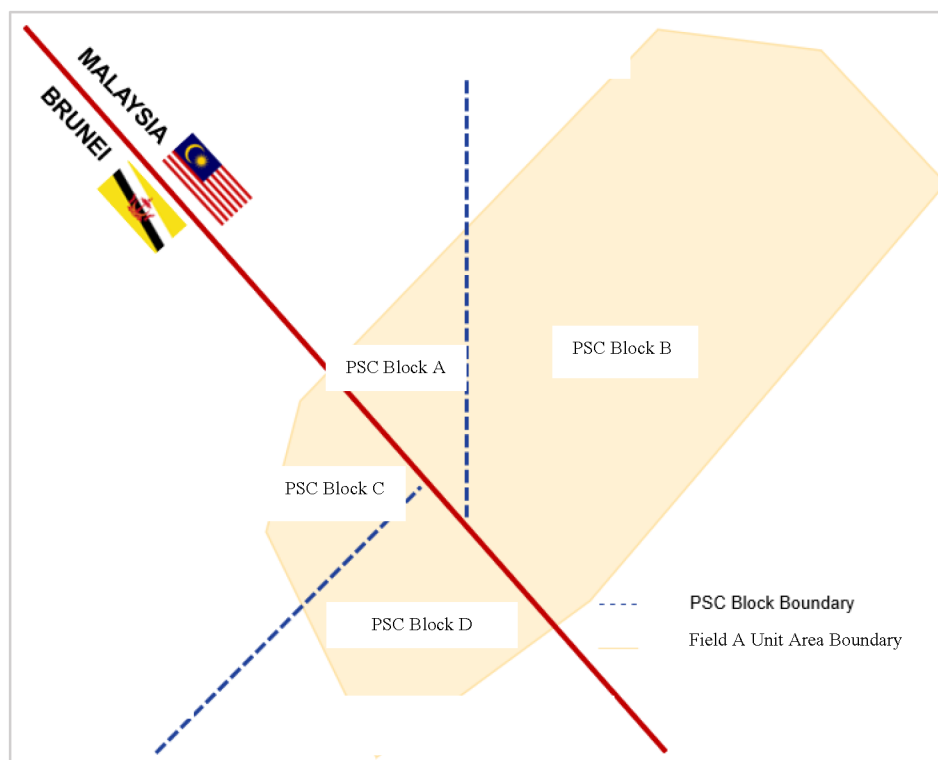


Figure 1—Straddling Field Map Illustration

Field A was initially discovered in 2003 and in year 2007 a phased development program was proposed to optimize the overall development of the Malaysia field, and this was approved by all joint venture partners and MPM. The decision was made to develop the Field A via subsea manifolds, flowlines and risers to Floating Processing System (FPS) with a design service life of 30 years. Oil will be exported through an 18" pipeline to the onshore terminal and production from the field commenced in October 2014 allowing production from the 6 oil producers. Phase 1 and Phase 2 production were expected to come off plateau in 2024 and continue declining thereafter. Development of Phase 3 and Phase 4 is important to sustain the production thereafter. The project has since evolved to include more than 20 wells across four phases, with development extending into Brunei.

Methodology

Field Development Concept

Field A Phase 4 Development is a growth driver for Malaysia Deepwater, where it is a continuation of additional subsea tiebacks to the FPS host facility, which enables new reserves to be developed efficiently and maximizes the usage of available FPS ullage. The overall development timing is influenced by

the anticipated ullage profile, expected scope maturation and Subsea Umbilical, Risers, and Flowlines (SURF) long lead item delivery duration. Infrastructure development was equally significant focuses on development of the south-west sector of the field where four-slot manifold was installed in Brunei waters and tied back to existing Malaysian facilities, integrating the two nations' offshore systems into a single operational network. This Field Development Plan (FDP) proposed 3 oil producers (OP) and 1 water injector (WI) wells subsea facility tie-ins to Drill Center C (DC-C) in Malaysia and Drill Center D (DC-D) in Brunei as shown in Figure 2.

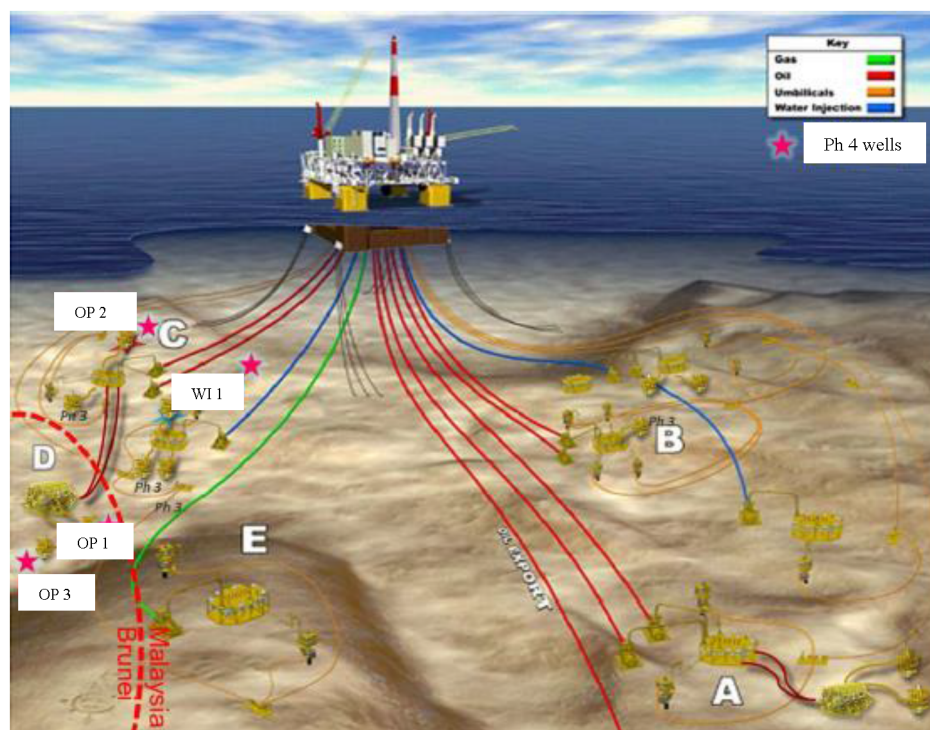


Figure 2—Field A Subsea Layout

Operational Polygon

An operational polygon is a geographically defined boundary (usually a polygon-shaped area) that sets the limits within which all project activities such as drilling, installation, and subsea operations will take place. It is typically agreed upon under Unitization Agreement (UA) or Joint Development Agreements (JDA) between two countries when a hydrocarbon reservoir straddles their maritime boundary.

The establishment of the operational polygon serves as a critical framework to streamline and govern offshore activities for this project. Its primary purpose is to clearly demarcate and confine all offshore execution operations including drilling, completion, SURF installation campaigns, as well as ongoing operations and maintenance of subsea wells within a precisely defined geographic area. This spatial boundary enables focused management of complex offshore tasks, ensuring operational clarity and safety.

A key function of the operational polygon is to grant direct access into Drill Centre-D, as authorized by the Department of Energy of Brunei Darussalam. This access facilitates efficient vessel and helicopter movements, allowing for optimized logistics by minimizing transit times between drill centers and reducing unnecessary delays. Vessels departed directly from Labuan, carrying fuel for bunkering and sailing to the rig location within the Operating Polygon. By enabling direct sailings from Labuan to DC-D Brunei as shown in Figure 3, the polygon supports smoother execution of campaigns and enhances resource utilization.

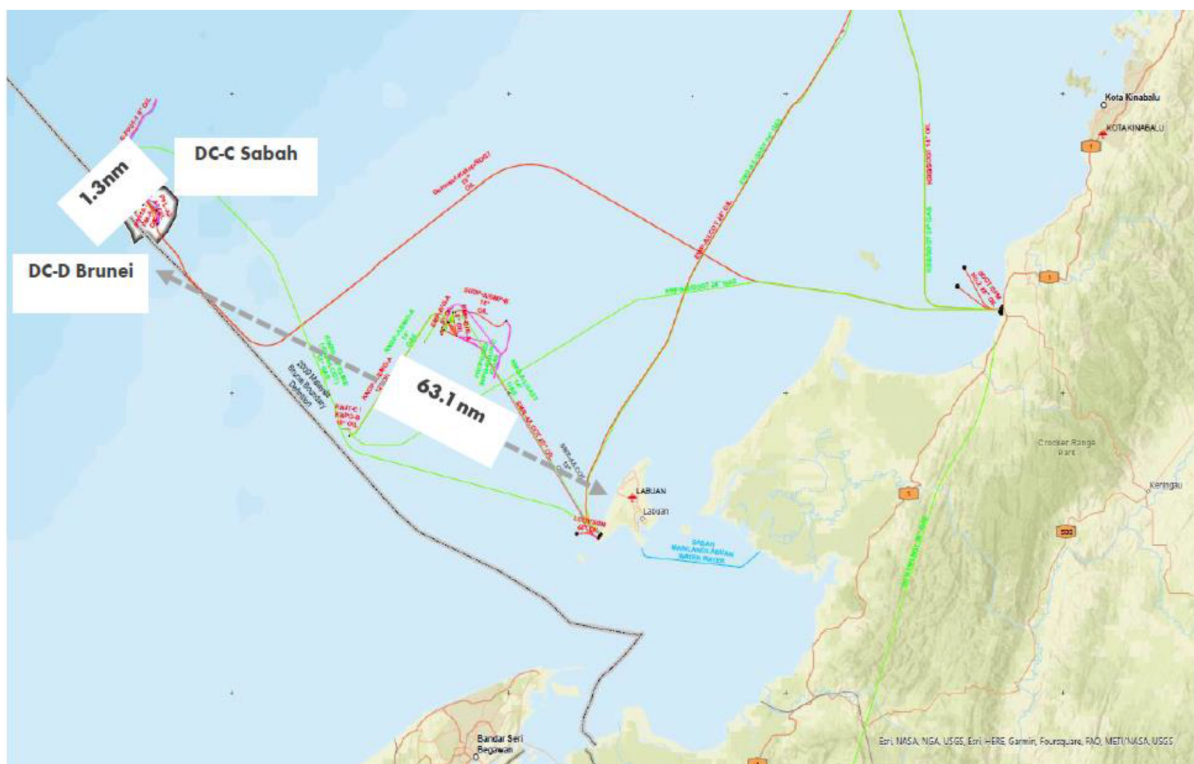


Figure 3—Routes without physical stopover

The operational polygon also ensures full regulatory compliance with the laws and regulations of both Brunei Darussalam and Malaysia. This dual adherence is essential given the cross-border nature of the project and underpins all activities within the operational boundary. It also provides a structured platform for enhanced cross-border coordination, simplifying ingress notifications, permitting processes, and emergency response planning between the two countries' authorities.

From a technical standpoint, the Operational Polygon mitigates challenges associated with drilling and installation activities by defining clear operational limits, thereby addressing potential technical restrictions. By doing so, it helps avoid the logistical complexities and additional costs that might arise from vessel or equipment movements outside the polygon.

Result and Discussion

Subsurface Key Uncertainties and Mitigations

The Field A development is characterized by significant geological complexity and subsurface uncertainty, requiring a carefully crafted drilling and completion strategy. The depositional environment dominated by turbidite lobes and distributary channels. The current subsurface plan targets two main reservoir intervals—Reservoir M and Reservoir N—via a planned sequence of producer and injector wells. Specifically, the campaign involves a producer-injector pair (OP 1 and WI 1) in the M reservoir and two producers (OP 2 and OP 3) in the N reservoir. These wells are designed to penetrate geologically challenging zones where seismic clarity is compromised, primarily due to the presence of a gas cloud above the Field A structure that reduces imaging quality.

This seismic limitation, compounded by limited well control in certain zones, introduces the risk of suboptimal well placement, which could impact ultimate recovery (UR). To mitigate this, the well planning team focused on selecting high-confidence targets that offer robust UR potential across a range of subsurface realizations. The strategy also includes an "appraisal while drilling" approach, in which M reservoir wells

are drilled first to gather data while drilling through the N reservoir. This allows for iterative optimization of the N well targets based on the latest information. Trajectory designs have been carefully engineered to accommodate depth uncertainty, with real-time data integration enabling continuous target refinement. From a technology perspective, the project leveraged advanced 4D Ocean Bottom Node (OBN) Seismic technology, with data collected in 2015, 2018, and 2023. Seismic velocity reprocessing was conducted to improve imaging of the subsurface structure, which provided valuable insights

The uncertainty in sand body thickness and reservoir quality remains a primary challenge in the Field A development. The field is characterized by a compensational stacking depositional frontier model, which results in complex and laterally variable sand distributions. To address these challenges, the wells are designed to be highly deviated to maximise reservoir exposure to increase productivity while addressing risk of lateral reservoir heterogeneity

Historical drilling data from previous development phases used to confirm the high heterogeneity of sand quality and thickness, with abrupt changes observed over relatively small distances. This heterogeneity introduces significant risk in both well placement and completion strategy, require robust subsurface integration and real-time decision-making.

A further technical concern involves the risk of early water or gas breakthrough, often attributed to high-permeability streaks or injection conformance issues. To manage this, reservoir surveillance programs leveraging well performance trends, tracer analysis, and dynamic modelling are implemented. These tools enable timely detection of breakthrough events and support proactive adjustments in injection and offtake strategies, as well as remediation planning where necessary.

Sand production risk also poses a considerable operational challenge, particularly in wells where downhole sand control mechanisms fail. Wells with standalone screen completions and a history of water production have demonstrated elevated risk, based on analogy from earlier development phases. In the current phase, a gravel pack completion was proposed for one of the three planned oil producers to enhance sand control reliability. Meanwhile, standalone screens were selected for the ERD wells to maximize completion feasibility and flexibility, given the reservoir uncertainty in thickness and net-to-gross (NTG) variability. The inflow area of these wells was designed to be maximized to minimize drawdown and reduce the risk of screen hot spotting. In addition, the operating envelope will be defined based on calculated screen velocities to ensure operations remain within erosional velocity limits.

All these challenges translated into well performance and surface facilities availability particularly produced water and gas handling systems which remain critical enablers to realize the full potential of the new wells. This risk is being actively mitigated through close coordination with the asset reliability, operations, and production planning teams, ensuring continuous improvement in system uptime and availability.

Deepwater Drilling Technology and SURF campaign

This development represents a landmark achievement in Malaysia's upstream oil and gas industry, overcoming a spectrum of complex technical and regulatory challenges. Located across international boundaries, the project contended with depleted reservoirs, shallow geohazards, poor seismic clarity due to gas cloud interference, and EERD requirements all within a stringent cross-border regulatory framework.

To address these hurdles, the project applied several advanced technologies such as managed pressure drilling (MPD), liner flotation, and novel cementing methods. Drilling a 6-kilometer EERD horizontal well across multiple hydrocarbon-bearing sands and depleted reservoirs presents significant operational complexities. Although such wells typically require detailed well analysis, the importance of robust contingency planning particularly for issues like stuck pipe cannot be underestimated. In the case of OP 1, two times cased hole sidetrack was necessary within the 9-5/8" liner due to unfavorable reservoir conditions encountered earlier. This non-conventional deployment of the 9-5/8" liner string is used to ensure reach of liner as intended; hence enabling the potential of tapping into reservoir at Brunei side without stepping too

Wellbore strengthening (WBS) formulations are designed to plug up to >1,500-micron fracture to enhance the fracture gradient of depleted sands. This enabled critical cementation (barrier) to be executed and avoided major downtime such as severe losses, stuck pipe and potential sidetracks. The WBS materials was tested through slot to assess the potential plugging risks of bottom hole assembly (BHA). Thixotropic gas-tight cement slurries were deployed to safely navigate shallow hazards and maintain well integrity

The SURF campaign was successfully completed on schedule despite operating during the monsoon season, effective mitigation strategies were implemented to maintain progress and safety. These included conducting interim mobilizations in sheltered waters rather than at the field itself and prioritizing the installation of large critical structures such as manifolds, suction piles, umbilical termination assemblies (UTAs), and mud mats during favorable weather windows. Operational efficiency was further enhanced by optimizing work scopes to run in parallel where safety permitted, for example initiating flowline and umbilical installations during interim mobilizations and performing offshore leak tests with back-seal tests concurrently with remotely operated vehicle (ROV) deployments. Additionally, the streamlining of mechanical completion processes and subsea as-built documentation facilitated simultaneous commissioning activities, minimizing overall project duration and improving integration between installation and commissioning phases. [Figure 5](#) and [6](#) shows pictures from pre installation and SURF campaign.

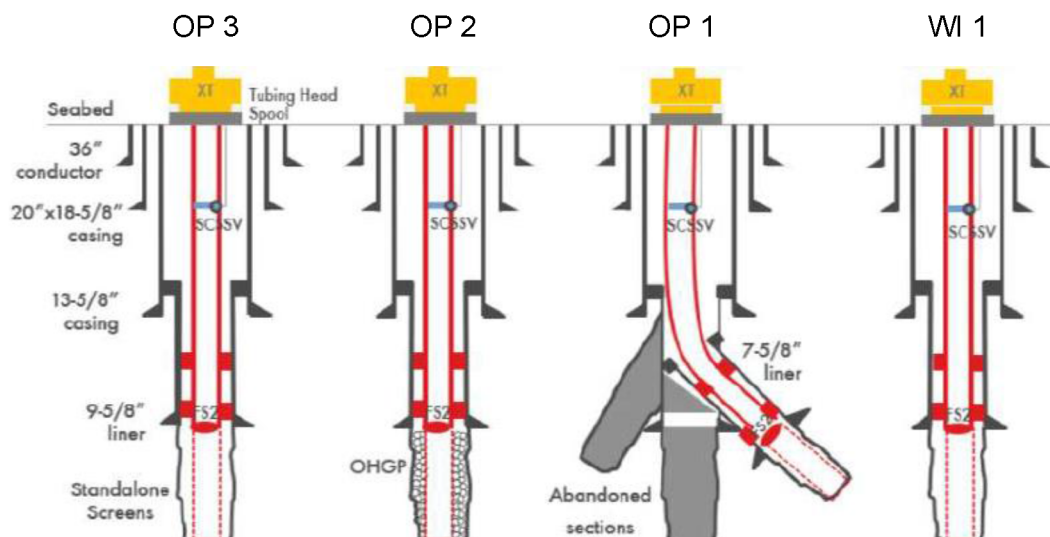


Figure 4—Adaptive strategy through secured sidetrack equipment for OP1 with alternate locations

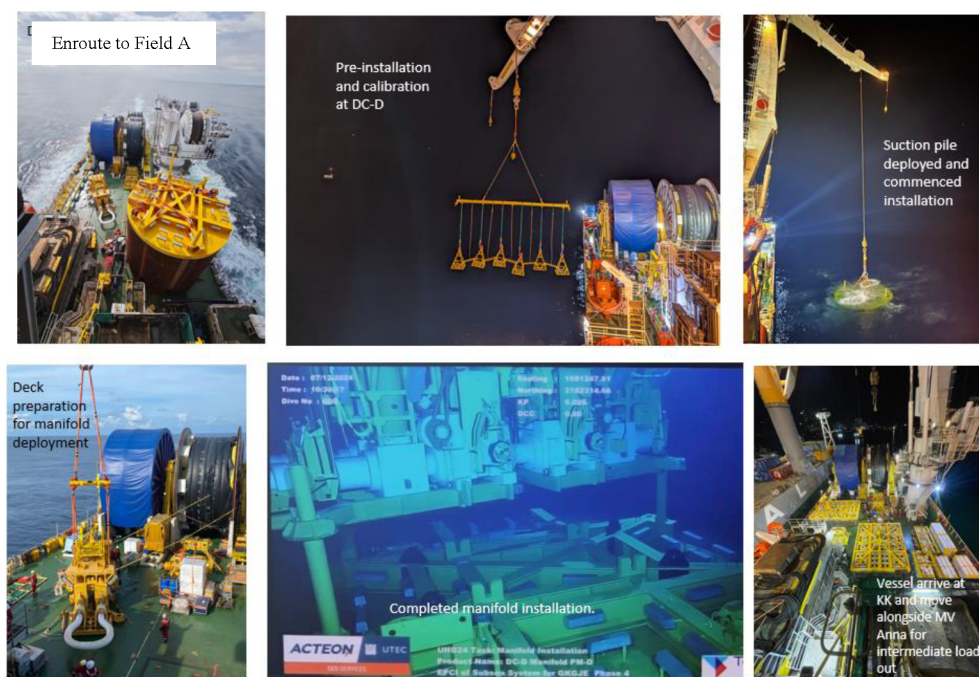


Figure 5—Pre-installation survey at DC-D and installation of manifold



Figure 6—Commencement of the Field A Phase 4 SURF Campaign in Nov 2024

Cross Border Execution

Cross-border oil and gas developments present unique opportunities and challenges, requiring a blend of technical expertise, diplomatic engagement, and regulatory alignment. Building on the exchange of letters that resolved long-standing territorial disputes, both nations have forged a partnership that balances national sovereignty with shared economic and environmental responsibilities. The partnership is anchored in clear governance structures, mutual trust, and aligned strategic goals, forming a strong foundation based on following key principles of collaboration.

- **Respecting Sovereignty and Territorial Integrity**

Both countries maintain respect for each other's sovereignty and maritime boundaries. The framework ensures that no disputes arise over ownership of natural resources, enabling stability and certainty for investors. The agreement facilitated peaceful cooperation in Block A-D

- **Joint Development and Resource Sharing**

Where maritime boundaries converge, joint development agreements allow for cooperative exploration, development, and extraction of hydrocarbons. Mechanisms such as Production Sharing Agreements (PSAs) ensure fair benefit distribution while avoiding infringement on territorial claims.

- **Realizing Mutual Economic Benefit**

The collaboration aims to deliver balanced economic returns for both nations through shared revenues, job creation, and local industry stimulation. Profits and operational benefits are equitably allocated, ensuring sustainable growth.

- **Upholding Environmental Protection**

Both countries adhere to stringent environmental rules, using sustainable exploration and production methods that reduce harm to the sea and surrounding nature. They also meet international and regional environmental requirements to protect wildlife and biodiversity.

- **Enhancing Legal and Regulatory Alignment**

Cross-border operations need the legal and regulatory frameworks of both countries to be aligned or harmonized. Malaysia and Brunei work together to align their oil and gas laws, including taxation, safety regulations, environmental impact assessments, and other requirements, to ensure smooth operations, reduce legal complexity, and meet international standards

- **Capacity Building and Knowledge Transfer**

The partnership goes beyond economic gains, fostering skills development, knowledge transfer, and capability building among local personnel to ensure a resilient and competent workforce.

The Malaysia-Brunei cross-border key principles of collaboration model has translated into strategic and operational benefits that extend well beyond fiscal returns. A comprehensive framework was established to govern the cross-border operation, detailing critical elements for stakeholder mapping, process opportunities, and regulatory considerations for both Malaysia and Brunei. Emergency readiness is strengthened through pre-agreed joint emergency response plans and tabletop drills, minimizing delays during crisis scenarios. Regulatory processes are streamlined, achieving up to three months' time savings and over 10% cost avoidance through harmonized approvals, reporting, and logistics. Operational polygons enabling seamless cross-border movement of personnel, vessels, and equipment while lowering emissions and reducing costs. Additionally, collaborative governance through close engagement with over 20 regulatory bodies accelerates decision-making, simplifies logistics, and ensures operational transparency.

In future, potential incentives supporting cross-border operations could include the Investment Tax Allowance (ITA), Reinvestment Allowance (RA), Accelerated Capital Allowance (ACA), and Production Sharing Contracts (PSC) with cost recovery terms. Collectively, these measures enhance investment attractiveness while making complex cross-border projects more financially viable and strategically appealing.

Conclusion and Novelty

The collaborative nature of Field A Phase 4 execution delivered far-reaching benefits for both Malaysia and Brunei. Strategically, it represents a landmark in bilateral cooperation, setting a precedent for peaceful resource sharing that might have remained untapped without joint development. Economically, the project generated significant fiscal uplift for both nations through shared production revenues, while avoiding the duplication of costly infrastructure. The collaboration not only generates long-term value for both nations but also creates a sustainable and competitive advantage for the oil and gas industry in the region, positioning

Malaysia and Brunei as leaders in cross-border offshore development. Regulatory stewardship by Malaysia Petroleum Management (MPM), in concert with Brunei's authorities, ensured that the venture met stringent technical, environmental, and operational standards, while delivering optimum value to all stakeholders.

Despite facing significant challenges including subsurface uncertainties, EERD trajectories, complex reservoir heterogeneity, and logistical constraints, the project overcame these through robust planning, adaptive drilling strategies, advanced seismic imaging, and innovative well construction strategy. Operational efficiency was further enhanced through the establishment of an operational polygon, harmonized regulatory processes, and streamlined cross-border logistics, resulting in reduced costs, faster approvals, and lower emissions. These achievements not only demonstrate technical excellence but also reinforce PETRONAS' aspiration to position

Malaysia as the deepwater hub for the region, capable of leading high-impact projects that deliver novel value across multiple countries while advancing the frontiers of deepwater oil and gas development. Operationally, it streamlined cross -border processes, creating a replicable model for future joint developments in the region. This project was built on transparent communication, mutual trust, and aligned national interests, transforming a technical challenge into a diplomatic and economic success.

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